



Institute of Information Technology

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Lab Report - 02

Lab Title: Linux File Commands (File Management & Directory Management)

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Course Title: Operating System Lab

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1. Objective

In this lab, we practiced frequently used Linux commands for file and directory management. The goal was to get comfortable with the Linux terminal by running real commands and observing what each one does. We learned how to list files, navigate directories, create and delete files, and view file contents.

2. Theory

Linux is a command-line-based operating system where most operations are done through a terminal. Unlike graphical interfaces, the terminal gives more control over the system. In this lab, we used the Ubuntu terminal to run various file management commands.

Linux treats everything as a file including directories. Commands like `ls`, `cd`, `mkdir`, `cat`, `touch`, and `rm` are fundamental tools every Linux user needs to know. These commands are simple but very powerful when combined together.

We also learned that Linux uses a tree-like file structure starting from the root (`/`). Each user has a home directory (e.g., `/home/shaon`) and this is where we worked throughout the lab.

3. Commands Used

<code>ls</code>	Lists all files and directories in the current folder.
<code>ls -al</code>	Shows a detailed list including hidden files (files starting with a dot).
<code>ls -lt</code>	Lists files sorted by modification time, newest first.
<code>cd <dir></code>	Changes the current directory to the specified one.
<code>cd -</code>	Goes back to the previously visited directory.
<code>cd ..</code>	Moves up one level to the parent directory.
<code>cd</code>	Goes directly to the home directory.
<code>pwd</code>	Prints the full path of the current working directory.
<code>mkdir <dir></code>	Creates a new directory with the given name.

<code>cat > <file></code>	Creates a new file and lets the user type content into it.
<code>echo "text" >> <file></code>	Appends a line of text to an existing file.
<code>more <file></code>	Shows the file content page by page.
<code>head <file></code>	Displays the first 10 lines of a file.
<code>tail <file></code>	Displays the last 10 lines of a file.
<code>touch <file></code>	Creates an empty file or updates the timestamp of an existing file.
<code>clear</code>	Clears the terminal screen.
<code>rm <file></code>	Deletes the specified file.
<code>rm -r <dir></code>	Deletes a directory along with everything inside it.
<code>rm -f <file></code>	Force deletes a file without asking for confirmation.
<code>rm -rf <dir></code>	Force deletes a directory and all its contents recursively.

4. Procedure

We opened the terminal on Ubuntu and ran each command one by one. For every command, we first ran it, then observed the output and noted what happened. Below is a step-by-step description of what we did.

4.1 ls Directory Listing

Lists all non-hidden files and folders in the current directory.

Command:

```
$ ls
```

Output:

```
Desktop  Documents  Downloads  Music  Pictures  Projects  Public  snap
Templates  Videos
```

We saw all the directories inside the home folder. Hidden files were not visible with plain ls.

4.2 ls -al Detailed Listing with Hidden Files

Shows a detailed list including file permissions, owner, size, date, and hidden files.

Command:

```
$ ls -al
```

Output:

```
total 88
drwxr-x--- 17 shaon shaon 4096 Apr 22 20:31 .
-rw----- 1 shaon shaon 1510 Apr 22 20:29 .bash_history
... (and more files including hidden ones starting with .)
```

We noticed that hidden files like `.bashrc`, `.bash_history`, and `.profile` appeared here but not in the plain `ls` output.

4.3 `ls -lt` Sort by Modification Time

Lists files and directories sorted by last modified time, newest first.

Command:

```
$ ls -lt
```

Output:

```
total 40
drwxr-xr-x 2 shaon shaon 4096 Apr 22 20:31 Downloads
drwxrwxr-x 2 shaon shaon 4096 Apr 15 14:45 Projects
...
```

The Downloads folder appeared at the top because it was modified most recently. This is useful when you need to quickly find recently changed files.

4.4 `cd` Change Directory

Changes the working directory to the specified folder.

```
$ cd Documents
$ cd -          (back to previous)
$ cd ..        (go up one level)
$ cd           (go to home)
```

Output:

```
/home/shaon
```

We moved into Documents, then used `cd -` to go back to where we were. The `cd ..` moved us up one level. Just typing `cd` brought us back home instantly.

4.5 `pwd` Print Working Directory

Displays the full path of the current directory.

Command:

```
$ pwd
```

Output:

```
/home/shaon
```

This was helpful when we were not sure which directory we were currently in. It always shows the absolute path.

4.6 mkdir Make Directory

Creates a new directory with the given name.

Command:

```
$ mkdir Projects
```

Output:

```
mkdir: cannot create directory 'Projects': File exists (if already exists)
```

Otherwise, no output – directory is created silently.

When we tried to create Projects again, it gave an error saying it already exists. Linux does not overwrite existing directories with mkdir.

4.7 cat > file Create File with Input

Creates a new file and allows typing content directly from keyboard. Press Ctrl+D to save.

Command:

```
$ cat > notes.txt
Hello, This is test file,
created by Shaon Khan
Who are dreamer
want success and try to work hard.
[Ctrl+D]
```

Output:

```
(File notes.txt is created with the typed content)
```

We typed multiple lines and pressed Ctrl+D to finish. The file was saved immediately. We then used cat notes.txt to verify the content.

4.8 echo >> Append to File

Appends a new line of text to an existing file without overwriting it.

Command:

```
$ echo "This is new line which appends at the last of previous text"
>> notes.txt
```

Output:

```
(No output – text is appended silently to notes.txt)
```

After running this, we used `more notes.txt` to see that the new text was added at the end of the existing content.

4.9 more View File Page by Page

Displays file content one screen at a time. Press Space to continue, q to quit.

Command:

```
$ more notes.txt
```

Output:

```
Hello, This is test file,  
created by Shaon Khan  
Who are dreamer  
want success and try to work hard.This is new line...
```

The `more` command was useful when the file was long. It showed content page by page, which was easier to read than `cat`.

4.10 head First 10 Lines

Prints the first 10 lines of the file.

Command:

```
$ head notes.txt
```

Output:

```
Hello, This is test file,  
created by Shaon Khan  
Who are dreamer  
want success and try to work hard.This is new line...  
(first 10 lines shown)
```

The `head` command is handy when a file is very large and we only need to check what is at the beginning.

4.11 tail Last 10 Lines

Prints the last 10 lines of the file.

Command:

```
$ tail notes.txt
```

Output:

```
10
```

```
11
12
13
14
14
15
16
17
18
```

The tail command showed the final 10 lines. We used it after adding many numbered lines to the file using echo -e.

4.12 touch Create Empty File

Creates an empty file. If the file already exists, it updates its timestamp.

Command:

```
$ touch hello.txt
```

Output:

```
(No output - hello.txt is created empty)
```

We used ls after touch to confirm the file was created. The file was empty, which we verified with cat hello.txt.

4.13 rm Remove File

Deletes the specified file permanently.

Command:

```
$ rm hello.txt
```

Output:

```
(No output - file is deleted)
```

After running rm, we did ls again and confirmed hello.txt was gone. There is no trash or undo in Linux, so we need to be careful with rm.

4.14 rm -r Remove Directory

Deletes a directory and all its contents recursively.

Command:

```
$ rm -r Projects
```

Output:

```
rm: cannot remove 'Projects': No such file or directory (if already removed)
Otherwise removes silently.
```

We first tried to remove Projects while we were inside it, which gave an error. After going up (cd), it worked correctly.

4.15 rm -f Force Delete File

Force deletes a file without any confirmation prompt.

Command:

```
$ rm -f notes.txt
```

Output:

```
(No output — file deleted without asking)
```

Unlike rm, this does not prompt at all. It is useful when you want to remove write-protected files quickly.

4.16 rm -rf Force Delete Directory

Force deletes a directory and all its contents without any prompts.

Command:

```
$ rm -rf Pro
```

Output:

```
(No output — directory and all its contents deleted)
```

This command is very powerful and dangerous. We created a test directory Pro, then removed it completely in one command.

4.17 clear Clear the Terminal

Clears all previous output from the terminal screen.

Command:

```
$ clear
```

Output:

```
(Terminal screen is cleared)
```

After running many commands, the screen was cluttered. The clear command made it clean again. The command history was still accessible with the up arrow key.

5. Discussion

In this lab, we ran 18 different Linux commands and observed how each one works. A few things stood out to us during the session.

First, we noticed that `ls` and `ls -al` show very different results. The plain `ls` command only shows visible files, while `ls -al` reveals hidden configuration files like `.bashrc` and `.bash_history`. These hidden files are important for system settings.

Second, we learned that the `>>` operator appends text to a file without overwriting it, while `>` overwrites. This is an important distinction — using the wrong one can delete existing content.

Third, the `rm` command is permanent in Linux. There is no recycle bin. So we had to be careful, especially with `rm -rf`, which deletes everything in a directory silently.

We also noticed that `echo -e` allows us to use escape characters like `\n` for newlines. When we added multiple numbered lines to `notes.txt`, we could then verify them using `head` and `tail` commands, which are very efficient for checking large files.

Overall, this lab helped us understand how the Linux terminal handles files and directories. The commands are simple but require careful usage.

6. Conclusion

We successfully completed Lab 2 of the Operating System Lab course. In this session, we practiced all the fundamental Linux file and directory management commands. We created files, wrote content, appended data, viewed files in different ways, and removed files and directories.

Through this lab, we gained practical experience with the Linux terminal. We now feel more confident using the command line for everyday file operations. The knowledge gained here will be helpful for more advanced Linux and OS topics in future labs.

7. Lab Screenshots

The following screenshots were captured during the lab session and show the actual terminal output for the commands we ran.

```
shaon@shaon:~$ ls
Desktop Documents Downloads Music Pictures Projects Public snap Templates Videos
shaon@shaon:~$ ls -al
total 88
drwxr-x--- 17 shaon shaon 4096 Apr 22 20:31 .
drwxr-xr-x  3 root  root  4096 Apr  8 18:31 ..
-rw-----  1 shaon shaon 1510 Apr 22 20:29 .bash_history
-rw-r--r--  1 shaon shaon  220 Apr  8 18:31 .bash_logout
-rw-r--r--  1 shaon shaon 3771 Apr  8 18:31 .bashrc
drwx----- 16 shaon shaon 4096 Apr  8 18:55 .cache
drwx----- 16 shaon shaon 4096 Apr 15 14:50 .config
drwxr-xr-x  2 shaon shaon 4096 Apr  8 18:40 Desktop
drwxr-xr-x  2 shaon shaon 4096 Apr  8 18:40 Documents
drwxr-xr-x  2 shaon shaon 4096 Apr 22 20:31 Downloads
drwx-----  2 shaon shaon 4096 Apr  8 18:44 .gnupg
drwx-----  3 shaon shaon 4096 Apr  8 18:40 .local
drwxr-xr-x  2 shaon shaon 4096 Apr  8 18:40 Music
-rw-r--r--  1 shaon shaon  361 Apr 15 14:42 .pam_environment
drwxr-xr-x  3 shaon shaon 4096 Apr  8 18:40 Pictures
-rw-r--r--  1 shaon shaon  807 Apr  8 18:31 .profile
drwxrwxr-x  2 shaon shaon 4096 Apr 15 14:45 Projects
drwxr-xr-x  2 shaon shaon 4096 Apr  8 18:40 Public
drwx-----  4 shaon shaon 4096 Apr  8 18:50 snap
drwx-----  2 shaon shaon 4096 Apr  8 18:44 .ssh
drwxr-xr-x  2 shaon shaon 4096 Apr  8 18:40 Templates
drwxr-xr-x  2 shaon shaon 4096 Apr  8 18:40 Videos
shaon@shaon:~$ ls -lt
total 40
drwxr-xr-x 2 shaon shaon 4096 Apr 22 20:31 Downloads
drwxrwxr-x 2 shaon shaon 4096 Apr 15 14:45 Projects
drwx----- 4 shaon shaon 4096 Apr  8 18:50 snap
drwxr-xr-x 3 shaon shaon 4096 Apr  8 18:40 Pictures
drwxr-xr-x 2 shaon shaon 4096 Apr  8 18:40 Desktop
drwxr-xr-x 2 shaon shaon 4096 Apr  8 18:40 Documents
drwxr-xr-x 2 shaon shaon 4096 Apr  8 18:40 Music
```

Screenshot 1: ls and ls -al commands showing directory listing and hidden files

```

shaon@shaon:~$ ls -lt
total 40
drwxr-xr-x 2 shaon shaon 4096 Apr 22 20:31 Downloads
drwxrwxr-x 2 shaon shaon 4096 Apr 15 14:45 Projects
drwx----- 4 shaon shaon 4096 Apr 8 18:50 snap
drwxr-xr-x 3 shaon shaon 4096 Apr 8 18:40 Pictures
drwxr-xr-x 2 shaon shaon 4096 Apr 8 18:40 Desktop
drwxr-xr-x 2 shaon shaon 4096 Apr 8 18:40 Documents
drwxr-xr-x 2 shaon shaon 4096 Apr 8 18:40 Music
drwxr-xr-x 2 shaon shaon 4096 Apr 8 18:40 Public
drwxr-xr-x 2 shaon shaon 4096 Apr 8 18:40 Templates
drwxr-xr-x 2 shaon shaon 4096 Apr 8 18:40 Videos
shaon@shaon:~$ cd Documents
shaon@shaon:~/Documents$ ls
shaon@shaon:~/Documents$ cd -
/home/shaon
shaon@shaon:~$ cd ..
shaon@shaon:/home$ cd
shaon@shaon:~$

```

Screenshot 2: `ls -lt` and `cd` commands sorted listing and directory navigation

```

shaon@shaon:~$ ls
Desktop Documents Downloads Music Pictures Projects Public snap Templates Videos
shaon@shaon:~$ ls -al
total 88
drwxr-x--- 17 shaon shaon 4096 Apr 22 20:31 .
drwxr-xr-x 3 root root 4096 Apr 8 18:31 ..
-rw----- 1 shaon shaon 1510 Apr 22 20:29 .bash_history
-rw-r--r-- 1 shaon shaon 220 Apr 8 18:31 .bash_logout
-rw-r--r-- 1 shaon shaon 3771 Apr 8 18:31 .bashrc
drwx----- 16 shaon shaon 4096 Apr 8 18:55 .cache
drwx----- 16 shaon shaon 4096 Apr 15 14:50 .config
drwxr-xr-x 2 shaon shaon 4096 Apr 8 18:40 Desktop
drwxr-xr-x 2 shaon shaon 4096 Apr 8 18:40 Documents
drwxr-xr-x 2 shaon shaon 4096 Apr 22 20:31 Downloads
drwx----- 2 shaon shaon 4096 Apr 8 18:44 .gnupg
drwx----- 3 shaon shaon 4096 Apr 8 18:40 .local
drwxr-xr-x 2 shaon shaon 4096 Apr 8 18:40 Music
-rw-r--r-- 1 shaon shaon 361 Apr 15 14:42 .pam_environment
drwxr-xr-x 3 shaon shaon 4096 Apr 8 18:40 Pictures
-rw-r--r-- 1 shaon shaon 807 Apr 8 18:31 .profile
drwxrwxr-x 2 shaon shaon 4096 Apr 15 14:45 Projects
drwxr-xr-x 2 shaon shaon 4096 Apr 8 18:40 Public
drwx----- 4 shaon shaon 4096 Apr 8 18:50 snap
drwx----- 2 shaon shaon 4096 Apr 8 18:44 .ssh
drwxr-xr-x 2 shaon shaon 4096 Apr 8 18:40 Templates
drwxr-xr-x 2 shaon shaon 4096 Apr 8 18:40 Videos
shaon@shaon:~$ ls -lt
total 40
drwxr-xr-x 2 shaon shaon 4096 Apr 22 20:31 Downloads
drwxrwxr-x 2 shaon shaon 4096 Apr 15 14:45 Projects
drwx----- 4 shaon shaon 4096 Apr 8 18:50 snap
drwxr-xr-x 3 shaon shaon 4096 Apr 8 18:40 Pictures
drwxr-xr-x 2 shaon shaon 4096 Apr 8 18:40 Desktop
drwxr-xr-x 2 shaon shaon 4096 Apr 8 18:40 Documents
drwxr-xr-x 2 shaon shaon 4096 Apr 8 18:40 Music

```

Screenshot 3: `ls -al` and `ls -lt` command outputs

```

drwxr-xr-x 2 shaon shaon 4096 Apr  8 18:40 Public
drwxr-xr-x 2 shaon shaon 4096 Apr  8 18:40 Templates
drwxr-xr-x 2 shaon shaon 4096 Apr  8 18:40 Videos
shaon@shaon:~$ cd Documents
shaon@shaon:~/Documents$ ls
shaon@shaon:~/Documents$ cd -
/home/shaon
shaon@shaon:~$ cd ..
shaon@shaon:/home$ cd
shaon@shaon:~$ pwd
/home/shaon
shaon@shaon:~$ mkdir Projects
mkdir: cannot create directory 'Projects': File exists
shaon@shaon:~$ mkdir Projects
shaon@shaon:~$ ls
Desktop Documents Downloads Music Pictures Projects Public snap Templates Videos
shaon@shaon:~$ cd Projects
shaon@shaon:~/Projects$ ls
shaon@shaon:~/Projects$ cat > notes.txt
Hello, This is test file,
created by Shaon Khan
Who are dreamer
want success and try to work hard.shaon@shaon:~/Projects$
shaon@shaon:~/Projects$ cat notes.txt
Hello, This is test file,
created by Shaon Khan
Who are dreamer
want success and try to work hard.shaon@shaon:~/Projects$
shaon@shaon:~/Projects$ echo "This is new line which appends at the last of previous text" >> notes.
txt

```

Screenshot 4: mkdir, cd, cat > notes.txt creating directory and writing file

```

shaon@shaon:~/Projects$ more notes.txt
Hello, This is test file,
created by Shaon Khan
Who are dreamer
want success and try to work hard.This is new line which appends at the last of previous text
shaon@shaon:~/Projects$ echo

shaon@shaon:~/Projects$ echo "This is
> new
> line
> which
> add
> at
> the
> last
> of
> the
> previous
> line
> seriously
> trust
> me
> .
> Our
> mam
> teaches
> us
> ok?" >> notes.txt

```

Screenshot 5: more notes.txt and echo append using multi-line input

```

shaon@shaon:~/Projects$ more notes.txt
Hello, This is test file,
created by Shaon Khan
Who are dreamer
want success and try to work hard.This is new line which appends at the last of previous text
This is
new
line
which
add
at
the
last
of
the
previous
line
seriously
trust
me
.
Our
mam
teaches
us
ok?
shaon@shaon:~/Projects$ echo -e"Line 1\n 2 \n 3 \n 3\n 4 \n 5 \n 6 \n 7 \n 8 \n 9 \n 10 \n 11 \n 12
\n 13 \n 14 \n 14 \n 15 \n 16 \n 17 \n 18" >> notes.txt

```

Screenshot 6: more notes.txt showing appended content, and echo -e command

```

shaon@shaon:~/Projects$ echo -e"Line 1\n 2 \n 3 \n 3\n 4 \n 5 \n 6 \n 7 \n 8 \n 9 \n 10 \n 11 \n 12
\n 13 \n 14 \n 14 \n 15 \n 16 \n 17 \n 18" >> notes.txt
shaon@shaon:~/Projects$ more notes.txt
Hello, This is test file,
created by Shaon Khan
Who are dreamer
want success and try to work hard.This is new line which appends at the last of previous text
This is
new
line
which
add
at
the
last
of
the
previous
line
seriously
trust
me
.
Our
mam
teaches
us
ok?
-eLine 1\n 2 \n 3 \n 3\n 4 \n 5 \n 6 \n 7 \n 8 \n 9 \n 10 \n 11 \n 12 \n 13 \n 14 \n 14 \n 15 \n 16
\n 17 \n 18
shaon@shaon:~/Projects$ echo -e "Line 1\n 2 \n 3 \n 3\n 4 \n 5 \n 6 \n 7 \n 8 \n 9 \n 10 \n 11 \n 12
\n 13 \n 14 \n 14 \n 15 \n 16 \n 17 \n 18" >> notes.txt

```

Screenshot 7: echo -e output with multiple numbered lines added to notes.txt

```
shaon@shaon:~/Projects$ more notes.txt
Hello, This is test file,
created by Shaon Khan
Who are dreamer
want success and try to work hard.This is new line which appends at the last of previous text
This is
new
line
which
add
at
the
last
of
the
previous
line
seriously
trust
me
.
Our
mam
teaches
us
ok?
-eLine 1\n 2 \n 3 \n 3\n 4 \n 5 \n 6 \n 7 \n 8 \n 9 \n 10 \n 11 \n 12 \n 13 \n 14 \n 14 \n 15 \n 16
\n 17 \n 18
Line 1
2
3
3
4
5
6
```

Screenshot 8: more notes.txt showing all content including numbered lines

```
7
8
9
10
11
12
13
14
14
15
16
17
18
shaon@shaon:~/Projects$ head notes.txt
Hello, This is test file,
created by Shaon Khan
Who are dreamer
want success and try to work hard.This is new line which appends at the last of previous text
This is
new
line
which
add
at
shaon@shaon:~/Projects$ tail notes.txt
10
11
12
13
14
14
15
16
17
18
```

Screenshot 9: Continuation of notes.txt, head and tail commands

```
10
shaon@shaon:~/Projects$ touch hello.txt
shaon@shaon:~/Projects$ ls
hello.txt  notes.txt
shaon@shaon:~/Projects$
```

Screenshot 10: touch hello.txt creating an empty file

```
shaon@shaon:~/Projects$ touch hello.txt
shaon@shaon:~/Projects$ ls
hello.txt  notes.txt
shaon@shaon:~/Projects$ rm hello.txt
shaon@shaon:~/Projects$ ls
notes.txt
shaon@shaon:~/Projects$ rm -r Projects
rm: cannot remove 'Projects': No such file or directory
shaon@shaon:~/Projects$ cd
shaon@shaon:~$ rm -r Projects
shaon@shaon:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
shaon@shaon:~$ touch notes.txt
shaon@shaon:~$ ls
Desktop  Documents  Downloads  Music  notes.txt  Pictures  Public  snap  Templates  Videos
shaon@shaon:~$ rm -f notes.txt
shaon@shaon:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
shaon@shaon:~$ mkdir Pro
shaon@shaon:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Pro  Public  snap  Templates  Videos
shaon@shaon:~$ rm -rf Pro
shaon@shaon:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
shaon@shaon:~$ clear
```

Screenshot 11: rm, rm -r, rm -f, rm -rf, and clear commands

```
shaon@shaon:~$
```

Screenshot 12: Terminal after clear command — clean screen